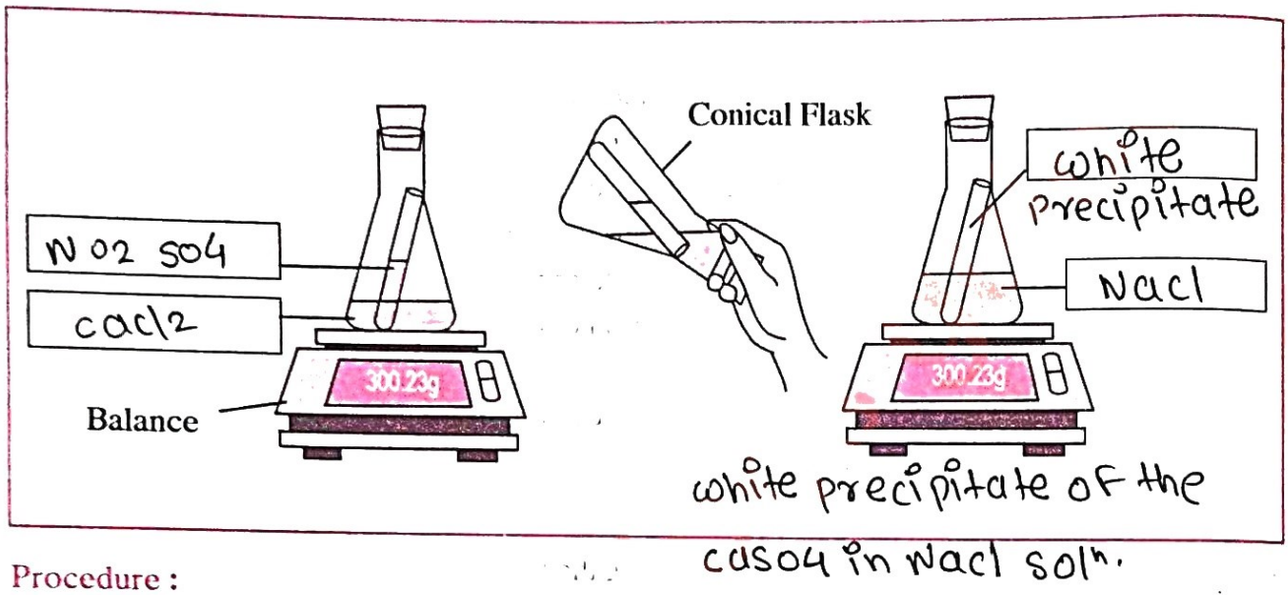
Experiment No. 3

Title : To verify the law of Chemical Combination.

Apparatus : Conical Flask, Test tube, Rubber cork, Balance etc .

Chemicals : Solutions of Calcium Chloride ($CaCl_2$), Sodium Sulphate (Na_2SO_4).

Figure :



Procedure :

1. Take a solution of calcium chloride ($CaCl_2$) in a conical flask.
2. Take a solution of sodium sulphate (Na_2SO_4) in a test tube.
3. Tie a thread of test tube and release it in the conical flask.
4. Make the conical flask airtight by putting a rubber cork.
5. Weigh the resulting assembly together with the balance.
6. Tilt the conical flask and mix the solutions in the conical flask and test tube in each other.
7. Weigh the assembly again.
8. Repeat the procedure 1 to 6 at least three times.
9. Compare the weights.

Observation :

Sr.No	Total mass of the assembly before reaction (g)	Total mass of the assembly after the reaction (g)
1	300.23 g.	300.23g
2	84.02g.	84.02g.
3	120.50g.	120.50g.

Inference :

To study laco of chemical combination with the apportus chemical such as cacl2, nasO4.

Multiple Choice Questions

1. The precipitate formed in the above experiment is of .....
   A. NaCl ✓B. $CaSO_4$ C. Na_2Cl_2 D. Na_2Ca
2. The concept .....is proved in the above experiment.
   A . Law of constant proportion B. Law of conservation of energy
   ✓C. Law of conservation of mass D. Mole concept
3. In the above experiment the weight of the substances is ..... the sum of the weight of original substances.
   ✓A. the same as B. less than C. more than D. double

Questions

1. Which other chemicals can be used for verifying the law of chemical combination?

1) The chemical such an, HC2, COCa3, NaSO4
H+CI → HCI Hydrochoric acid

2) $CaCO_3 + 2HCl \rightarrow CO_2 + H_2O$ Hydrochloric acid.

2. How is the law of conservation of matter verified in the experiment of calcium chloride and sodium sulphate?

Total mass of the substance is contant after the chemical reaction form this experiment it is the veaified that the laco of conversation matters of calcium chloride and sodium sulphide.

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